

4.5

MH1200 Linear Algebra I.

The 3rd Quiz, 2023.

- Please write down your **name** and **matriculation number** at the top. The duration is **15 minutes**. There is **one** question worth **5** marks.
- You may write on every part of this document and also use extra paper if you need. To get full marks please write down all of your working.

Problem 1:

(5 marks.)

Consider the following subset of $\mathbb{R}^3$:

$$S = \{a(1, 0, 0) + b(0, 2, 0) + c(-1, -1, 0) \in \mathbb{R}^3; a, b, c \in \mathbb{R}, \underline{a \leq 0, b \leq 0, c \leq 0}\}.$$

- Give three different vectors lying in S .
- State the axioms of a subspace.
- For each of the axioms of a subspace, prove that it is either satisfied or not satisfied by S . Is S a subspace of $\mathbb{R}^3$?

a) $(0, 0, 0)$, ~~$(0, -1, 0)$~~ , $(1, -1, 0)$

b) 1. The set contains the zero vector.

2. The set is closed under vector addition

3. The set is closed under vector multiplication

c) Axiom 1:

The zero vector, $\vec{0} = (0, 0, 0)$ can be found in S .

When $a=0, b=0, c=0$, it satisfies the condition that $a \leq 0, b \leq 0, c \leq 0$

$$\therefore 0(1, 0, 0) + 0(0, 2, 0) + 0(-1, -1, 0) = (0, 0, 0)$$

So S satisfies the first axiom

Axiom 2:

Let $\vec{v}_1$ and $\vec{v}_2$ be vectors lying in S .

$$\vec{v}_1 = a_1(1, 0, 0) + b_1(0, 2, 0) + c_1(-1, -1, 0)$$

~~$$\vec{v}_1 = a_1(1, 0, 0) + b_1(0, 2, 0) + c_1(-1, -1, 0)$$~~

~~$$\vec{v}_2 = a_2(1, 0, 0) + b_2(0, 2, 0) + c_2(-1, -1, 0)$$~~

$$\vec{v}_2 = a_2(1, 0, 0) + b_2(0, 2, 0) + c_2(-1, -1, 0)$$

$$\vec{v}_1 + \vec{v}_2 = a_1(1, 0, 0) + b_1(0, 2, 0) + c_1(-1, -1, 0) + a_2(1, 0, 0) + b_2(0, 2, 0) + c_2(-1, -1, 0)$$

$$= (a_1 + a_2)(1, 0, 0) + (b_1 + b_2)(0, 2, 0) + (c_1 + c_2)(-1, -1, 0)$$

Since $a \leq 0, b \leq 0, c \leq 0$, then $a_1 + a_2 \leq 0, b_1 + b_2 \leq 0, c_1 + c_2 \leq 0$.

So, since $(a_1 + a_2), (b_1 + b_2)$ and $(c_1 + c_2)$ are all less than or equal to zero, they satisfy the condition, and hence S is closed under vector addition.

c) Axiom 3:

S is not closed under vector multiplication

let $\vec{v}$ be a vector in S .

let $d \in \mathbb{R}$.

$$\vec{v} = a(0,0) + b(0,2,0) + c(-1,-1,0)$$

$$d\vec{v} = d(a(0,0) + b(0,2,0) + c(-1,-1,0))$$

$$\text{let } a = -1, b = -1, c = -1, d = -1$$

$$d\vec{v} = (0,1,0).$$

However

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

To check if it lies in the subspace

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The first row is inconsistent, so it does not have a solution.

This is not satisfied under vector multiplication

$$(a-c, 2b-c, 0)$$

$$S = \{a(1,0,0), b(0,2,0), c(-1,-1,0) \in \mathbb{R}^3, a, b, c \in \mathbb{R}, a \geq 0, b \geq 0, c \leq 0\}$$

$$\text{let } u = a-c, u \in \mathbb{R}, v = 2b-c, v \in \mathbb{R}$$

$$\text{is equivalent to } S = \{(u, v, 0) \in \mathbb{R}^3, u, v \in \mathbb{R}\}$$

$$\text{Axiom 2: closure under addition } S = \{a-c, 2b-c, 0\}$$

$$\text{Assume } (a_1-c_1, 2b_1-c_1, 0), (a_2-c_2, 2b_2-c_2, 0) \in S \text{ and}$$

$$a_1, \dots, c_2 \in \mathbb{R}.$$

$$(a_1-c_1, 2b_1-c_1, 0) + (a_2-c_2, 2b_2-c_2, 0) = (a_1+a_2-c_1-c_2, \dots)$$

$$= (a_1+a_2)(1,0,0) + (b_1+b_2)(0,2,0) + (c_1+c_2)(-1,-1,0)$$

$$\therefore a_1, a_2 \leq 0 \Rightarrow a_1+a_2 \leq 0$$

$$b_1, b_2 \leq 0 \Rightarrow b_1+b_2 \leq 0$$

$$c_1, c_2 \leq 0 \Rightarrow c_1+c_2 \leq 0$$

Thus S is closed under addition

Axiom 3: closed under multiplication

$$(a-c, 2b-c, 0) \quad u \in \mathbb{R}, u \text{ is a scalar}$$

$$u(a-c, 2b-c, 0)$$

Case 1: Suppose $u \geq 0$

$$ua(1,0,0) + ub(0,2,0) + uc(-1,-1,0)$$

$$ua \geq 0, ub \geq 0, uc \leq 0.$$

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she confused also

Case 2: Suppose $u < 0$

$$-uc(1,0,0) - uc(0,1,0) + ua(-1,0,0) + ub(0,-1,0)$$